International Institute of Information Technology, Hyderabad

(Deemed to be University)

S24SC1.308 : The Universe Across Scales

Question cum Answer Booklet

Mid-Semester Examination

Total Marks: 30 Duration : 1.5 hours

Roll No. : 2022111004 Program : CSD Date : 1 MARCH 2025

Room No. : H-301 Seat No. : A1 Invigilator Signature :

Marks Table (to be filled in by the examiner)

Question No.	Marks							Examiner
1	3	3						
2	3							
3	2							
4	2							
5	2							
6	3							
7	5							
8	3							MS
9	4							MS

Instructions to the students:

1. Place your ID card on the desk for verification by the invigilator.
2. Reading materials, such as books, and electronic gadgets, like tablets, computers, mobile phones or calculators, are not allowed inside the classroom.
3. Borrowing writing materials or calculators from other students during the exam is prohibited. If any student is found indulging in malpractice or copying during the exam, the student will be given an 'F' grade for the course.

1. Consider a linear chain of N Ising spins ($\sigma_i = \pm 1$) with a nearest neighbours coupling K and an external magnetic field H. Assume periodic boundary condition.

(a) Write down the Hamiltonian and associated partition function. Explain the terms.

(b) Using the transfer matrix, show that (only for two spins) the trace of the transfer matrix can be mapped with the associated partition function. **[3 + 3 = 6]**

a) $H_{\{\sigma_1, \sigma_2 \dots \sigma_N\}} = -K\left(\sum_{\langle \sigma_i, \sigma_j \rangle} \sigma_i \sigma_j\right) - H\left(\sum_{\langle \sigma_i \rangle} \sigma_i\right)$

Hamiltonian for a state with spins $\sigma_1, \sigma_2 \dots \sigma_N$

nearest neighbour coupling. sum of all adjacent pairs.

individual spins $\sum_{\langle \sigma_i \rangle} \sigma_i = \sum_{i=1}^{N} \sigma_i$

3

$\sum_{\langle \sigma_i, \sigma_j \rangle} \sigma_i \sigma_j = \left(\sum_{i=1}^{N-1} \sigma_i \sigma_{i+1} + \sigma_N \sigma_1\right)$

$K \to$ coupling strength

$H \to$ external magnetic field.

(Partition function in next page)

b) Transfer matrix $P = \begin{bmatrix} e^{\beta(K+H)} & e^{-\beta K} \\ e^{-\beta K} & e^{\beta(K-H)} \end{bmatrix}$ Periodic boundary condition.

partition function $Z = \text{trace}(P^2) \leftarrow$ as 2 spins.

$$P^2 = \begin{bmatrix} e^{\beta(K+H)} & e^{-\beta K} \\ e^{-\beta K} & e^{\beta(K-H)} \end{bmatrix} \begin{bmatrix} e^{\beta(K+H)} & e^{-\beta K} \\ e^{-\beta K} & e^{\beta(K-H)} \end{bmatrix} = \begin{bmatrix} e^{2\beta(K+H)} + e^{-2\beta K} & - \\ - & e^{-2\beta K} + e^{2\beta(K-H)} \end{bmatrix}$$

$$\text{trace}(P^2) = e^{2\beta(K+H)} + e^{-2\beta K} + e^{-2\beta K} + e^{2\beta(K-H)}$$

$$Z = e^{2\beta(K+H)} + e^{2\beta(K-H)} + 2e^{-2\beta K} \quad \text{——— (1)}$$

3

consider $Z = \sum_{\{\sigma_i\}} e^{-\beta H_i}$ sum over all possible states $\to \uparrow\uparrow \quad \uparrow\downarrow \quad \downarrow\uparrow \quad \downarrow\downarrow$

$$= \sum_{\{\sigma_i\}} e^{\beta K(\sum \sigma_i \sigma_j) + \beta H(\sum \sigma_i)}$$

$$= \underset{\uparrow\uparrow}{e^{\beta K(1+1) + \beta H(2)}} + \underset{\uparrow\downarrow}{e^{\beta K(-1-1) + \beta H(0)}} + \underset{\downarrow\uparrow}{e^{\beta K(-1-1) + \beta H(0)}} + \underset{\downarrow\downarrow}{e^{\beta K(1+1) + \beta H(-1-1)}}$$

$$= e^{2\beta K + 2\beta H} + 2e^{-2\beta K} + e^{2\beta K - 2\beta H}$$

$$Z = e^{2\beta(K+H)} + e^{2\beta(K-H)} + 2e^{-2\beta K} \quad \text{——— (2)}$$

(1) and (2) are the same equations.

— Hence proved.

perfect

a) Partition function :

$$Z = \sum_{\{i\}} e^{-\beta H_i}$$

↓ Partition function.

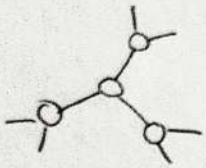
{all possible states.

H_i → Hamiltonian for state i

$$\beta = \frac{1}{k_B T}$$

2. Let, each site of a Bethe lattice has z neighbouring sites. Let g denotes the distance from a centre site.

Prove that the ratio of surface sites (i.e. number of sites at the surface when $g \to \infty$) to the total number sites does not depend on the distance g. [3]



Each site has z neighbouring sites

starting from arbitary center (going outwards), each node branches off to 2 more nodes.

generation = 0 nodes = 1

$g = 1$ $n = 2$

$g = 2$ $n = 2(2-1)$

$g = g$ $n = 2(2-1)^{g-1}$

} surface sites

At arbitary generation g.

no of surface sites $= 2(2-1)^{g-1}$

no. of total sites $= 1 + 2 + 2(2-1) + 2(2-1)^2 + \dots 2(2-1)^{g-1}$

$= 1 + 2\left[1 + (2-1) + (2-1)^2 + \dots (2-1)^{g-1}\right]$

↳ GP with $a=1$, $r = 2-1$, $n = g$

sum $= \dfrac{a(r^n - 1)}{r-1}$

$= 1 + 2\left[\dfrac{(2-1)^g - 1}{2-2}\right]$ $= \dfrac{(2-1)^g - 1}{2-2}$

$$\text{ratio} = \frac{2(2-1)^{g-1}(2-2)}{2(2-1)^g - \cancel{2} + \cancel{2} - 2} = \frac{2(2-1)^{g-1}(2-2)}{2(2-1)^g - 2}$$

divide N and D by $(2-1)^{g-1} \cdot 2$

$$= \frac{(2-2)}{(2-1) - \frac{2}{2(2-1)^{g-1}}}$$

$$\lim_{g \to \infty} \text{ratio} = \frac{(2-2)}{(2-1) - 0} = \frac{2-2}{2-1}$$ — Ans.

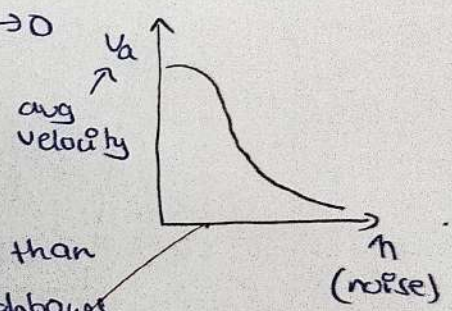
↳ no dependance on g.

3

3. (a) What type of interaction governs particle alignment in the Vicsek Model?
(b) What happens when the noise parameter in the Vicsek Model is increased? (It is not necessary to explain the governing equations of motion.) [1 + 1 = 2]

a) Local interactions with all neighbours with a circle of radius R_0 contribute to the particle alignment.
These interactions need not be physical connections but just an influence.

b) if noise parameter (η) is increased, then the disorder in the system increases. as disorder increases, particle velocities become random thus avg velocity $\to 0$



high $\eta \Rightarrow$ It implies that each update in the particle's direction is more dependant on random zero mean gaussian noise rather than the local interactions with the neighbours.

2

4. **Prove that fractal dimension of the Sierpinski triangle is less than 2 but greater than 1.** [2]

construction of Sierpinski Triangle.

Each full triangle is divided into 4 parts by making the length 1/2 and the middle piece is discarded.

∴ The progess of size of the shape at each generation:

$r: \; 1 \to \frac{1}{2} \to \frac{1}{4} \to \cdots \frac{1}{2^g}$

The no of triangles required to make the bigger shape for each scale changes as

$N(r): \; 1 \to 3 \to 9 \to \cdots 3^g$

The factal dimension:

$$d_f = \frac{\log(N(r))}{\log\left(\frac{1}{r}\right)} = \frac{\log(3^g)}{\log\left(1/\left(\frac{1}{2^g}\right)\right)} = \frac{\log(3^g)}{\log(2^g)} = \frac{g\log 3}{g\log 2}$$

$$= \frac{\log 3}{\log 2}$$

also $\log 2 < \log 3 < \log 4$ (monotonically increasing func

$$\therefore \frac{\log 2}{\log 2} < \frac{\log 3}{\log 2} < \frac{\log 4}{\log 2}$$

$$1 < d_f < \underbrace{\frac{\log 2^2}{\log 2}}_{2}$$

2

$\therefore 1 < d_f < 2$

Exactly $d_f = 1.58$

— Ans.

5. (a) If each pair of n nodes is connected independently with probability p in Erdos-Renyi graph (the degree distribution follows a Binomial distribution), what is the expected average degree of the graph? Additionally, what is the critical probability required for the emergence of a giant connected component? (A detailed analytical argument is not necessary.) [2]

→ Average ~~outgoing~~ degree:

each node ~~can~~ can be connected to $(n-1)$ other nodes

if prob of each connection is p

The distribution of k nodes being the degree is

$$\binom{n-1}{k} p^k (1-p)^{n-k}$$ → binomial distribution $(N-1, p)$

Binom(n,p) has mean $= np$.

∴ Expected value of ~~outg~~ degree = mean of Binom$(n-1, p)$

$$\langle k \rangle = (n-1)p.$$

— Ans

→ when the outgoing degree $\langle k \rangle$ is 1, there is going to be one component of order $n^{2/3}$

as $\langle k \rangle > 1$, a giant cluster begins forming which grows in size as $\langle k \rangle$ increased.

∴ giant cluster emerges when $\langle k \rangle = (n-1)p > 1$

$$\Rightarrow p > \frac{1}{n-1}$$

$$\therefore P_c = \frac{1}{n-1}$$

— Ans.

2

6. (a) Assume there is no force to accelerate or decelerate the relative motion between any two galaxies. If the distance of separation between two galaxies at present is r then show that the age of the universe is $t_0 = 1/H_0$. Calculate the numerical value of t_0. $1Mpc = 3.262 \times 10^6$ light years, $H_0 = 72 km/s/MPc$, and light speed $c = 3 \times 10^8 m/s$ [2+1]

Let the galaxies be A and B

separated by distance r at the present epoch. (t_0)

separated by distance 0 at big bang ($t=0$)

∴ The speed at which the galaxies are moving apart

$$V = \frac{\text{distance}}{\text{time}} = \frac{r-0}{t_0-0} = \frac{r}{t_0} \quad \text{— (1)}$$

Hubbles observation :

Light from distant galaxies appear redshifted. (2)

further apart – more redshift. $z \propto d$ (distance) b/w galaxy and observer.

if redshift is attributed to doppler effect,

then $z \propto V$

Hubbles law : $V = H_0 d$. — (2)

↳ Hubbles constant.

using (1) and (2) ($r = d$)

$$V = \frac{r}{t_0} = H_0 r \Rightarrow t_0 = \frac{1}{H_0}$$

— Ans.

$$t_0 = \frac{Mpc \times s}{72\ km}$$

$1 Mpc = 3.262 \times 10^6$ ly.

1 ly = dist travelled by light in a year.

$= 9.786 \times 10^{14}\ m/s$

$= 9.786 \times 10^{14}\ \frac{m}{s} \cdot year$

$= \text{speed}(c) \times \text{time}$

$= 3 \times 10^8\ \frac{m}{s} \times 1\ year$

$$= \frac{9.786 \times 10^{14}\ m \cdot year}{72 \times 10^3\ m}$$

$= 1.36 \times 10^{10}$ years ~ 13 billion years.

— Ans.

$\sim 1.36 \times 10^{10} \times 365 \times 24 \times 3600$ sec

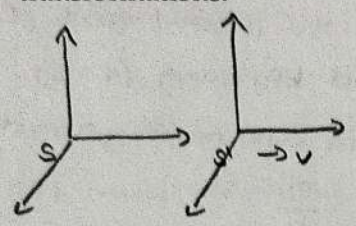
$= 4.29 \times 10^7 \times 10^{17}$ sec

$\sim 4.3 \times 10^{24}$ sec

— Ans.

/3

7. (a) Consider two infinitesimally close events in spacetime such that in a frame S the coordinates are (t, x, y, z) and $(t + dt, x + dx, y + dy, z + dz)$, and in S' frame the coordinates are (t', x', y', z') and $(t' + dt', x' + dx', y' + dy', z' + dz')$. The S' moves with velocity v along the positive x direction with respect to S and their coordinates are parallel to each other. Prove that the spacetime interval (also known as the line element, or the metric) is invariant under Lorentz transformations. [5]



Events $\mathcal{E}_1$ and $\mathcal{E}_2$

Spacetime Interval $(ds)^2 = -(dt)^2 + (dx)^2 + (dy)^2 + (dz)^2$

assuming $c = 1$

LT: $\begin{bmatrix} t' \\ x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} t \\ x \\ y \\ z \end{bmatrix}$ $\quad \gamma = \frac{1}{\sqrt{1-\beta^2}} \quad \beta = \frac{v}{c}$

this transformation also ~~stes~~ works in small scale $dx, \ldots$

$$\begin{bmatrix} dt' \\ dx' \\ dy' \\ dz' \end{bmatrix} = \Lambda \begin{bmatrix} dt \\ dx \\ dy \\ dz \end{bmatrix} \rightarrow \begin{aligned} dt' &= \gamma dt - \gamma\beta dx \\ dx' &= -\gamma\beta dt + \gamma dx \\ dy' &= dy \\ dz' &= dz \end{aligned}$$

Space time Interval in S' frame $(ds')^2 = -(dt')^2 + (dx')^2 + (dy')^2 + (dz')^2$

$$(ds')^2 = -(\gamma dt - \gamma\beta dx)^2 + (-\gamma\beta dt + \gamma dx)^2 + dy^2 + dz^2$$

$$= -\gamma^2 dt^2 - \gamma^2\beta^2 dx^2 + \cancel{2\gamma^2\beta dx\, dt}$$

$$+ \gamma^2 dx^2 + \gamma^2\beta^2 dt^2 - \cancel{2\gamma^2\beta dx\, dt} + dy^2 + dz^2$$

$$= dt^2(-\gamma^2 + \gamma^2\beta^2) + dx^2(\gamma^2 - \gamma^2\beta^2) + dy^2 + dz^2$$

$$= -(dt)^2 \gamma^2 (1-\beta^2) + dx^2 \gamma^2 (1-\beta^2) + dy^2 + dz^2$$

$$\gamma^2(1-\beta^2) = \left(\frac{1}{\sqrt{1-\beta^2}}\right)^2 (1-\beta^2) = \frac{(1-\beta^2)}{(1-\beta^2)} = 1$$

$$= -(dt)^2 + (dx)^2 + (dy)^2 + (dz)^2$$

$$= ds^2$$

↳ space time Interval in S frame.

$\therefore ds^2$ is Invariant under LT

— Ans.

8. (a) As measured by an observer in S frame, a flashbulb goes off at $x = 100$km, $y = 10$km, $z = 1$km, at $t = 5 \times 10^{-4}$s. What are the coordinates x', y', z', t' of this event as determined by a second observer in S' frame that is moving relative to S at $-0.8c$ along the $x - x'$ axis. You are given $c = 3 \times 10^8$m/s. [3]

Given, event in S frame $(t, x, y, z) = (c \times 5 \times 10^{-4}\,s, 100\,km, 10\,km, 1\,km)$

S' frame moving relative to S at $-0.8c$ along x axis.

$v = -0.8c \rightarrow \beta = \frac{v}{c} = -0.8 \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-0.64}} = \frac{1}{0.6} = \frac{10}{6} = \frac{5}{3}$

using LT

$$\begin{bmatrix} ct' \\ x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} ct \\ x \\ y \\ z \end{bmatrix}$$

$\gamma = \frac{5}{3} = 1.66$

$\gamma\beta = -1.33 \Rightarrow -\gamma\beta = 1.33$

15×10^4

event in S

(150 km, 100 km, 10 km, 1 km)

$$= \begin{bmatrix} 1.66 & 1.33 & & \\ 1.33 & 1.66 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 150 \\ 100 \\ 10 \\ 1 \end{bmatrix} \text{units in km}$$

$$= \begin{bmatrix} 382 \\ 366 \\ 10 \\ 1 \end{bmatrix}$$

$\therefore ct' = 382\,km \Rightarrow t' = \frac{382 \times 10^3}{3 \times 10^8}$ m/s

$= 127 \times 10^{-5}$

$t' = 1.27 \times 10^{-3}$ s.

$x' = 366$ km

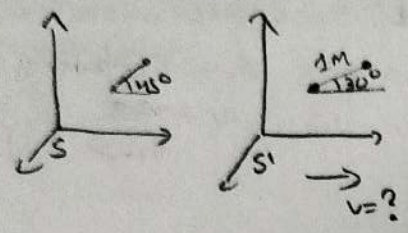
$y' = 10$ km $\quad z' = 1$ km.

Ans: $x' = 366$ km $\quad y' = 10$ km $\quad z' = 1$ km $\quad t' = 1.27 \times 10^{-3}$ s.

3

9. (a) A stick of one meter makes an angle of 30° with respect to the x'-axis in S' frame (the stick is rest in this frame). Seen from another frame S the stick makes an angle 45° with respect to its x axis. What is the relative velocity between the two frames. [4]

Assume relative velocity only along x axis:



S': The stick can be broken into two parts

$0.86m = \cos(30°)$ along x axis

$0.5m = \sin(30°)$ along y axis

S: The stick makes 45° with x axis

length along y wont change as motion is along x

$\therefore$ 0.5m along y axis

as $\theta = 45°$ $\Rightarrow$ 0.5m along x axis as well.

$\therefore$ horizontal stick of length 0.86m in S' appears as 0.5 m in S.

Say the horizontal stick in S' has ends x_1' and x_2' measurements at t_1' and t_2' $\quad x_2' - x_1' = 0.86$

$x_2 - x_1 = 0.5$

stick in S $\rightarrow$ x_1 and x_2

t $\quad$ t $\rightarrow$ simultaneous measurement as stick is moving wrt S.

using LT transform:

$x' = \gamma t - \gamma\beta x$

$\Rightarrow x_1' = \gamma t - \gamma\beta x_1$ and $x_2' = \gamma t - \gamma\beta x_2$

$x_2' - x_1' = \gamma t - \gamma\beta x_2 - \gamma t + \gamma\beta x_1$

$= -\gamma\beta(x_2 - x_1)$

$x' = -\gamma\beta t + \gamma x$ $\quad \Rightarrow x_1' = -\gamma\beta t + \gamma x_1$ and $x_2' = -\gamma\beta t + \gamma x_2$

$x_2' - x_1' = \gamma(x_2 - x_1)$ $\quad \Rightarrow \gamma = \dfrac{0.86}{0.5} = \dfrac{1}{\sqrt{1-\beta^2}}$

$\Rightarrow 1 - \beta^2 = \left(\dfrac{0.5}{0.86}\right)^2 = (0.58)^2 = 0.336$

$\Rightarrow \beta^2 = 1 - 0.336 = 0.664$.

$\Rightarrow \beta = 0.815 = \dfrac{v}{c}$ $\quad \Rightarrow v = 0.815c$

$= 0.815 \times 3 \times 10^8$ m/s

$= 2.4 \times 10^8$ m/s.

Ans: S' moving 2.4×10^8 m/s along +x axis of S frame.

4